PROJECT REPORT

# SMART RANDOM QUIZ APPLICATION

A Beginner-Level Python Project

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| **Programming Language** | Python |
| **Development Environment** | Visual Studio Code |

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# ABSTRACT

The Smart Random Quiz Application is a beginner-level console-based project developed using Python. It allows a user to choose one of three difficulty levels: Easy, Medium, or Hard. Based on the selected level, the program randomly selects five questions from the corresponding question bank. The user enters an option (A, B, C, or D), and the application checks the answer immediately. At the end of the quiz, the program displays the score, percentage, and a performance message. A replay option allows the user to start another round or exit the application.

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# 1. INTRODUCTION

Quizzes are commonly used to test knowledge and improve learning. This project provides a simple interactive quiz that runs in the terminal. It is designed for beginners who are learning Python programming concepts such as variables, lists, dictionaries, loops, conditional statements, functions of the random module, and user input.

# 2. OBJECTIVES

The main objectives of this project are:

• To develop an interactive quiz application using Python.

• To provide Easy, Medium, and Hard difficulty levels.

• To display five randomly selected questions in each quiz round.

• To accept answers from the user and check them automatically.

• To calculate and display the score and percentage.

• To provide feedback based on the user's performance.

• To allow the user to play again or exit the application.

# 3. SOFTWARE AND HARDWARE REQUIREMENTS

## Software requirements

| Item | Requirement |
|---|---|
| Operating System | Windows / Linux / macOS |
| Programming Language | Python 3.x |
| Code Editor | Visual Studio Code |
| Python Extension | Microsoft Python extension for VS Code (recommended) |

## Hardware requirements

A computer or laptop capable of running Python, with a keyboard and sufficient memory and storage for the operating system and code editor.

# 4. FEATURES AND MODULES

**1. Difficulty selection:** The user selects Easy, Medium, or Hard before starting the quiz.

**2. Separate question banks:** Each difficulty level has its own list of questions, options, and correct answers.

**3. Random question selection:** The random.sample() method selects five different questions from the chosen question bank for each round.

**4. Answer validation:** The program compares the user's answer with the stored correct answer. The upper() method makes answer checking case-insensitive.

**5. Score calculation:** One point is added for each correct answer.

**6. Percentage and feedback:** The application calculates the percentage and displays a performance message.

**7. Replay option:** The while loop allows another round when the user enters yes. Any other response exits the program.

# 5. ALGORITHM

1. Start the program.

2. Display the application title and difficulty options.

3. Read the user's difficulty choice.

4. Choose the corresponding question bank. If the choice is invalid, use Easy by default.

5. Randomly select five questions from the chosen question bank.

6. Set the score to zero.

7. For each selected question, display the question and options.

8. Read the user's answer and compare it with the correct answer.

9. If the answer is correct, increase the score; otherwise, display the correct answer.

10. After five questions, calculate the percentage and display the result message.

11. Ask whether the user wants to play again.

12. If yes, repeat the quiz; otherwise, display a thank-you message and stop.

# 6. PROGRAM FLOW

**Start** → Display title → Select difficulty → Choose question bank → Randomly select 5 questions → Display question → Accept answer → Check answer → Update score → Repeat until 5 questions are completed → Calculate percentage → Display result → Ask “Play again?” → **Yes:** restart quiz / **No:** display thank-you message → **End**

The flow repeats from the beginning when the user enters yes. The selected difficulty can be changed for each new round.

# 7. SAMPLE OUTPUT

The following is an illustrative run of the application using Easy Level:

```
Welcome to Smart Random Quiz Application!

Select Difficulty Level:
1. Easy
2. Medium
3. Hard
Enter your choice (1/2/3): 1
You selected Easy Level!

Let's start the quiz!

What is 10 + 5?
A. 10
B. 15
C. 20
D. 25
Enter your answer (A/B/C/D): b
Correct answer!

Quiz Completed!
Your final score is: 5 / 5
Your percentage is: 100.0 %
Excellent! Perfect Score!

Do you want to play again? (yes/no): no

Thank you for playing!
```

Note: The actual question order and selected questions may differ because selection is random.

# 8. TESTING

| Test | Input / Action | Expected result |
|---|---|---|
| Difficulty selection | Enter 1, 2, or 3 | Corresponding question bank is selected |
| Random selection | Start a quiz round | Five different questions are selected |
| Correct answer | Enter the correct option | Correct-answer message and score increment |
| Wrong answer | Enter an incorrect option | Wrong-answer message and correct option shown |
| Result calculation | Complete five questions | Score, percentage, and feedback displayed |
| Replay | Enter yes | A new round starts |
| Exit | Enter no | Thank-you message and program ends |

# 9. CONCLUSION

The Smart Random Quiz Application was developed successfully using Python. It provides three difficulty levels, random question selection, automatic answer checking, score and percentage calculation, performance feedback, and a replay option. The project demonstrates the practical use of Python lists, dictionaries, loops, conditional statements, input handling, and the random module. It is suitable as a beginner-level programming project.

# FUTURE ENHANCEMENTS

• Add more questions to each difficulty level.

• Add categories such as Science, Mathematics, General Knowledge, and Programming.

• Add a timer for each question.

• Store player names and scores in a file or database.

• Develop a graphical user interface using Tkinter or a web-based interface.

# 10. SOURCE CODE

The complete source code below matches the console-based version developed and tested in Visual Studio Code.

```
import random

easy_questions = [
    {
        "question": "What is the capital of India?",
        "options": ["A. Mumbai", "B. Delhi", "C. Chennai", "D. Hyderabad"],
        "answer": "B"
    },
    {
        "question": "How many days are there in a week?",
        "options": ["A. 5", "B. 6", "C. 7", "D. 8"],
        "answer": "C"
    },
    {
        "question": "What is 10 + 5?",
        "options": ["A. 10", "B. 15", "C. 20", "D. 25"],
        "answer": "B"
    },
    {
        "question": "Which is the national animal of India?",
        "options": ["A. Lion", "B. Elephant", "C. Tiger", "D. Deer"],
        "answer": "C"
    },
    {
        "question": "How many months are there in a year?",
        "options": ["A. 10", "B. 11", "C. 12", "D. 13"],
        "answer": "C"
    },
    {
        "question": "Which planet is known as the Red Planet?",
        "options": ["A. Earth", "B. Mars", "C. Jupiter", "D. Venus"],
        "answer": "B"
    },
    {
        "question": "What is 5 * 5?",
        "options": ["A. 10", "B. 15", "C. 20", "D. 25"],
        "answer": "D"
    }
]

medium_questions = [
    {
        "question": "Which language is used for AI?",
        "options": ["A. Python", "B. HTML", "C. CSS", "D. SQL"],
        "answer": "A"
    },
    {
        "question": "Which symbol is used for comments in Python?",

        "options": ["A. //", "B. #", "C. <!--", "D. **"],
        "answer": "B"
    },
    {
        "question": "Which file extension is used for Python files?",
        "options": ["A. .html", "B. .py", "C. .css", "D. .java"],
        "answer": "B"
    },
    {
```

```
        "question": "Which device is used to type text?",
        "options": ["A. Mouse", "B. Keyboard", "C. Monitor", "D. Printer"],
        "answer": "B"
    },
    {
        "question": "Which part is called the brain of a computer?",
        "options": ["A. CPU", "B. Mouse", "C. Keyboard", "D. Monitor"],
        "answer": "A"
    },
    {
        "question": "What is the square of 6?",
        "options": ["A. 12", "B. 24", "C. 30", "D. 36"],
        "answer": "D"
    },
    {
        "question": "What is 100 / 10?",
        "options": ["A. 5", "B. 10", "C. 15", "D. 20"],
        "answer": "B"
    }
]

hard_questions = [
    {
        "question": "What is the output of len('Python')?",
        "options": ["A. 5", "B. 6", "C. 7", "D. 8"],
        "answer": "B"
    },
    {
        "question": "Which Python data type is immutable?",
        "options": ["A. List", "B. Dictionary", "C. Set", "D. Tuple"],
        "answer": "D"
    },
    {
        "question": "Which keyword is used to define a function in Python?",
        "options": ["A. function", "B. def", "C. fun", "D. define"],
        "answer": "B"
    },
    {
        "question": "What is the output of 2 ** 3?",

        "options": ["A. 5", "B. 6", "C. 8", "D. 9"],
        "answer": "C"
    },
    {
        "question": "Which data structure stores key-value pairs?",
        "options": ["A. List", "B. Tuple", "C. Dictionary", "D. String"],
        "answer": "C"
    },
    {
        "question": "Which keyword is used for exception handling?",
        "options": ["A. try", "B. repeat", "C. check", "D. test"],
        "answer": "A"
    },
    {
        "question": "What is the result of 10 % 3?",
        "options": ["A. 1", "B. 2", "C. 3", "D. 0"],
        "answer": "A"
    }
]

while True:

    print("\\nWelcome to Smart Random Quiz Application!")

    print("\\nSelect Difficulty Level:")
    print("1. Easy")
    print("2. Medium")
    print("3. Hard")

    level = input("Enter your choice (1/2/3): ")

    if level == "1":
        questions = easy_questions
        print("You selected Easy Level!")

    elif level == "2":
        questions = medium_questions
        print("You selected Medium Level!")

    elif level == "3":
        questions = hard_questions
        print("You selected Hard Level!")

    else:
        questions = easy_questions
        print("Invalid choice! Defaulting to Easy Level.")

    selected_questions = random.sample(questions, 5)


    print("\\nLet's start the quiz!")

    score = 0

    for q in selected_questions:
```

```
    print("\\n" + q["question"])

    for option in q["options"]:
        print(option)

    answer = input("Enter your answer (A/B/C/D): ").upper()

    if answer == q["answer"]:
        print("Correct answer!")
        score += 1
    else:
        print("Wrong answer!")
        print("Correct answer is:", q["answer"])

print("\\nQuiz Completed!")
print("Your final score is:", score, "/", len(selected_questions))

percentage = (score / len(selected_questions)) * 100

print("Your percentage is:", percentage, "%")

if percentage == 100:
    print("Excellent! Perfect Score!")
elif percentage >= 80:
    print("Very Good! Keep it up!")
elif percentage >= 60:
    print("Good! Keep Practicing!")
elif percentage >= 40:
    print("Nice Try! Practice more!")
else:
    print("Don't give up! Try again!")

play_again = input("\\nDo you want to play again? (yes/no): ").lower()

if play_again != "yes":
    print("\\nThank you for playing!")
    break
```